PARITY-REDUCED TRANSVERSE WRONSKIANS, HIGHER DIPOLE INVARIANTS, AND RIGOROUS CONTINUUM CERTIFICATION FOR RIEMANN'S Ξ-FUNCTION

THIJS MISSIAEN

ABSTRACT. We prove that the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ of Riemann's completed Ξ-function satisfies odd transverse parity $W(-d,t) = -W(d,t)$ with identically vanishing even derivatives $\partial_d^{2k}W(0,t) \equiv 0$, so that the parity-reduced Wronskian $\widetilde{W}(d,t) := W(d,t)/d$ removes the $d \to 0^+$ boundary-layer degeneracy in the Speiser–Hinkkanen–Lagarias positivity criterion. Our proof combines a division-free Wigner–hyperbolic integral representation governed by the monotone kernel $\mathrm{sinc}_{\mathrm{hyp}}(dy) = \frac{\sinh(dy)}{dy} \geq 1$, closed-form Cauchy–Riemann identities for the higher transverse dipole invariants $c_{2m-1}(t)$ and non-singular partial derivatives $\nabla\widetilde{W}$, and dual directed-rounding interval evaluators based on Euler–Maclaurin summation and Jacobi-theta quadrature. Applying two-dimensional interval mean-value enclosures, the integer argument principle, and continuum Lipschitz bounds with IEEE-754 rounding control, we establish strict positivity of the leading dipole moments $c_1(\gamma_k), c_3(\gamma_k), c_5(\gamma_k) > 0$ at the first ten non-trivial zeros and certify the absence of off-critical zeros across 4,987,101 continuum cells covering $[0, 0.50] \times [0, 55]$.

1. INTRODUCTION

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be Riemann's completed zeta function, and write $s = \frac{1}{2} + d + \mathrm{i}t$ with transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and spectral height $t \in \mathbb{R}$. Setting $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$, Riemann's entire function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$ admits the Fourier–Laplace integral representation [16, 14, 21]

$$\Xi(t - \mathrm{i}d) = 8\int_0^\infty \Phi(u)\cosh(2du)\cos(2tu)\,\mathrm{d}u + \mathrm{i}\,8\int_0^\infty \Phi(u)\sinh(2du)\sin(2tu)\,\mathrm{d}u, \tag{1.1}$$

where the rapidly decaying Jacobi–Riemann kernel $\Phi(u)$ is defined by

$$\Phi(u) := \sum_{n=1}^{\infty} \phi_n(u), \qquad \phi_n(u) := \left(2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}\right)\exp\left(-\pi n^2 \mathrm{e}^{4u}\right). \tag{1.2}$$

By the equivalence theorems of Speiser [18], Hinkkanen [6], and Lagarias [9], the absence of off-critical zeros of $\xi(s)$ in the half-plane $d > 0$ is intimately tied to the positivity of $\mathrm{Re}(\xi'(s)/\xi(s)) = W(d,t)/|\Xi(t - \mathrm{i}d)|^2$, where the *transverse Wronskian* is given via the Cauchy–Riemann equations ($\partial_d R = \partial_t I_{\mathrm{im}}$, $\partial_d I_{\mathrm{im}} = -\partial_t R$) by

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}|\Xi(t - \mathrm{i}d)|^2 = R\,\partial_d R + I_{\mathrm{im}}\,\partial_d I_{\mathrm{im}} = R\,\partial_t I_{\mathrm{im}} - I_{\mathrm{im}}\,\partial_t R. \tag{1.3}$$

A foundational obstacle in two-dimensional numerical and interval-arithmetic certification of the critical strip—highlighted by Spira [19, 20] and encountered whenever one attempts to enclose $W(d,t) > 0$ near $\sigma = \frac{1}{2}$—is the *boundary-layer degeneracy* as $d \to 0^+$: because $\Xi(t)$ is real-valued on the critical line $d = 0$, $I_{\mathrm{im}}(0,t) \equiv 0$ and $\partial_d R(0,t) \equiv 0$, forcing $W(0,t) \equiv 0$ identically for all $t \in \mathbb{R}$. Consequently, standard Lipschitz or interval boxes $B \subset [0, 0.50] \times [t_a, t_b]$ touching or approaching $d = 0$ inevitably contain zero in their enclosure of $W(B)$. The purpose of this article is to resolve this boundary-layer singularity analytically and to construct a division-free interval and continuum certification architecture across the critical strip.

Date: October 2026.

2020 *Mathematics Subject Classification.* Primary 11M06, 65G30; Secondary 30D15, 26D15, 11Y35.

Key words and phrases. Riemann Xi-function, parity-reduced transverse Wronskian, hyperbolic sinc kernel, Turan inequalities, Lehmer phenomenon, interval arithmetic, argument principle, continuum certification.

Theorem 1.1 (Parity-Reduced Wronskian, Dipole Hierarchy, and Continuum Certification)**.** *Let $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$ be defined by (1.1), and let $I(y,t) = 2\pi\mathcal{W}_{\widetilde{\Phi}}(y,2t)$ denote the Wigner correlation kernel of the rescaled wavepacket $\widetilde{\Phi}(w) := 2\Phi(w/2)$ established in [11]. Then the following structural and quantitative properties hold:*

(i) ***Odd parity reduction and division-free hyperbolic-sinc representation:*** *The transverse Wronskian satisfies $W(-d,t) = -W(d,t)$, so all even transverse derivatives vanish identically on the critical line: $\partial_d^{2k} W(0,t) \equiv 0$ for all $k \geq 0$. The parity-reduced Wronskian $\widetilde{W}(d,t) := W(d,t)/d$ extends to an even, real-analytic function on $\mathbb{R}^2$ given by*

$$\widetilde{W}(d,t) = \frac{1}{2}\int_0^\infty y^2\, \mathrm{sinc}_{\mathrm{hyp}}(dy)\, I(y,t)\,\mathrm{d}y, \qquad \mathrm{sinc}_{\mathrm{hyp}}(w) := \frac{\sinh w}{w} = \sum_{m=0}^{\infty} \frac{w^{2m}}{(2m+1)!} \geq 1, \tag{1.4}$$

with flat Neumann tangent $\partial_d \widetilde{W}(0,t) \equiv 0$, Lagarias–Turán boundary value $\widetilde{W}(0,t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t)$, and $\widetilde{W}(0,\gamma_k) = (\Xi'(\gamma_k))^2 > 0$ at every simple real zero γ_k.

(ii) ***Higher transverse dipole hierarchy and pure-square coalescence cascade:*** *The Taylor expansion $\widetilde{W}(d,t) = \sum_{m=1}^{\infty} c_{2m-1}(t)\, d^{2m-2}$ has coefficients*

$$\begin{aligned} c_{2m-1}(t) &:= \frac{1}{(2m-1)!}\partial_d^{2m-1} W(0,t) = \frac{1}{2\,(2m-1)!}\int_0^\infty y^{2m} I(y,t)\,\mathrm{d}y \\ &= m\sum_{j=0}^{2m} \frac{(-1)^{m+j}}{j!\,(2m-j)!}\,\Xi^{(j)}(t)\,\Xi^{(2m-j)}(t). \end{aligned} \tag{1.5}$$

If $t_0 \in \mathbb{R}$ is an m-fold coalesced zero ($\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$, $\Xi^{(m)}(t_0) \neq 0$), then $c_1(t_0) = \cdots = c_{2m-3}(t_0) = 0$ and the leading non-zero coefficient is the pure square $c_{2m-1}(t_0) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_0))^2 > 0$. At each of the first ten non-trivial zeros $\gamma_1, \ldots, \gamma_{10}$, directed-rounding interval arithmetic certifies $c_1(\gamma_k) > 0$, $c_3(\gamma_k) > 0$, and $c_5(\gamma_k) > 0$.

(iii) ***Division-free Cauchy–Riemann gradients and rigorous continuum certification:*** *In terms of the monotone kernel $\eta_{\mathrm{hyp}}(w) := \frac{w\cosh w - \sinh w}{w^3} \geq \frac{1}{3}$, $\widetilde{W}(d,t)$ and its gradient $\nabla\widetilde{W} = (\partial_d\widetilde{W}, \partial_t\widetilde{W})$ admit division-free integral formulas free of $0/0$ cancellation as $d \to 0$. Over the finite domain $[0, 0.50] \times [0, 55]$ (and macro-strip $[10^{-6}, 1.20] \times [0, 55]$), two-dimensional interval mean-value enclosures across 80 critical collar boxes, integer argument-principle enclosures $N_{\mathrm{zeros}}(\Omega_k) \in [-2.62 \times 10^{-14}, +2.62 \times 10^{-14}] \cap \mathbb{Z} = \{0\}$ on five macro-rectangles $\Omega_1, \ldots, \Omega_5$, and continuum Lipschitz bounds across 4,987,101 cells $C_{i,j}$ certify $\widetilde{W}(d,t) > 0$ everywhere on $[0, 0.50] \times [0, 55]$ and $N_{\mathrm{zeros}} = 0$ on $(0, 1.20] \times [0, 55]$.*

To fix the quantitative scale of Theorem 1.1, we collect the principal structural and certification constants established in Sections 3–5 in a single display:

$$\begin{aligned} \widetilde{W}(0,0) &= -\Xi(0)\Xi''(0) \approx 1.1420 \times 10^{-2}, & c_1(\gamma_1) &= (\Xi'(\gamma_1))^2 \approx 1.9119 \times 10^{-6}, \\ c_3(\gamma_1) &= \tfrac{1}{6}\partial_d^3 W(0,\gamma_1) \approx 2.5881 \times 10^{-7}, & c_5(\gamma_1) &= \tfrac{1}{120}\partial_d^5 W(0,\gamma_1) \approx 1.1553 \times 10^{-8}, \\ N_{\mathrm{cells}} &= 399 \times 12{,}499 = 4{,}987{,}101, & \min_{t\in[10,55]} \widehat{W}_{\mathrm{lower}}(C_{i,j}) &\approx +1.0175 \times 10^{3} > 0, \\ \min_{1\leq m\leq 80} \widetilde{W}(\mathbf{B}_m).a &\approx +1.5592 \times 10^{-12} > 0, & \max_{1\leq k\leq 5} |N_{\mathrm{zeros}}(\Omega_k)| &\leq 2.62 \times 10^{-14} < \tfrac{1}{2}. \end{aligned} \tag{1.6}$$

Table 1 situates the analytical and computational contributions of the present article relative to the classical literature.

1.1. **Proof overview.** The proof of Theorem 1.1 rests on three interconnected analytical and computational ingredients:

(1) *First structural ingredient (Odd transverse parity and hyperbolic-sinc reduction, Section 2):* Using the functional symmetry $\Xi(t+\mathrm{i}d) = \overline{\Xi(t-\mathrm{i}d)}$, we show that $|\Xi(t-\mathrm{i}d)|^2$ is even in d and $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ is odd in d, forcing $\partial_d^{2k}W(0,t) \equiv 0$ for all $k \geq 0$.

TABLE 1. Comparison between classical results in the literature and the novel analytical and computational contributions established in this work.

Classical Author(s)	**Established Classical Framework**	**Novel Contribution in This Work**
Spira (1968–1973) [19, 20], Hinkkanen (1997) [6], Lagarias (1999) [9]	Zero-free certification via $\mathrm{Re}(\xi'(s)/\xi(s)) > 0$; degeneracy $W(0,t) \equiv 0$ on the critical line $d = 0$.	Odd parity proof $\partial_d^{2k} W(0,t) \equiv 0$ and division-free reduction $\widetilde{W}(d,t) = W(d,t)/d$ with $\widetilde{W}(0,t) > 0$ and $\partial_d \widetilde{W}(0,t) \equiv 0$.
Karlin (1968) [8], Csordas–Norfolk–Varga (1986) [2], Griffin et al. (2019) [4]	Total positivity (TP_∞), Turán inequalities $c_1(t) = (\Xi')^2 - \Xi\Xi'' > 0$, and Jensen polynomials.	Higher transverse dipole hierarchy $c_{2m-1}(t)$, Wigner moment equivalence, and pure-square cascade $c_{2m-1}(t_0) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_0))^2$.
De Bruijn (1950) [3], Newman (1976) [12], Rodgers–Tao (2020) [17]	Backward heat-flow deformation $H_\Lambda(z)$ of the Jacobi-theta kernel $\Phi(u)$ and zero spacing.	Identification of $c_{2m-1}(t)$ with even Wigner moments $\int_0^\infty y^{2m} I(y,t)\,\mathrm{d}y$ and super-linear dipole repulsion $c_3(\gamma_k), c_5(\gamma_k) > 0$.
Turing (1953) [23], Lehmer (1956) [10], Higham (2002) [5], Platt (2017) [13]	1D critical-line zero isolation and error analysis of finite-precision floating-point arithmetic.	Dual directed-rounding interval evaluators (`mpmath.iv`), 2D Interval-MVT collar boxes, integer argument principle, and 2D continuum Lipschitz certification.

Factoring d explicitly inside the Wigner phase-space integral via $\mathrm{sinc}_{\mathrm{hyp}}(dy) = \frac{\sinh(dy)}{dy}$ produces

$$\widetilde{W}(d,t) = \frac{W(d,t)}{d} = \frac{1}{2}\int_0^\infty y^2\, \mathrm{sinc}_{\mathrm{hyp}}(dy)\, I(y,t)\,\mathrm{d}y, \qquad \partial_d \widetilde{W}(0,t) = 0, \tag{1.7}$$

which eliminates the $d \to 0^+$ boundary-layer singularity and links $\widetilde{W}(0,t)$ directly to the Lagarias–Turán quadratic form $(\Xi'(t))^2 - \Xi(t)\Xi''(t)$.

(2) *Second quantitative ingredient (Higher dipole hierarchy and non-singular Cauchy–Riemann gradients, Section 3):* Expanding $|\Xi(t - \mathrm{i}d)|^2 = \Xi(t - \mathrm{i}d)\Xi(t + \mathrm{i}d)$ via Leibniz's rule and the Cauchy–Riemann identities $\partial_d^n \Xi(t - \mathrm{i}d)|_{d=0} = (-\mathrm{i})^n \Xi^{(n)}(t)$ establishes the closed-form dipole coefficients (1.5), including the cubic and quintic invariants

$$\begin{aligned} c_3(t) &= \frac{1}{6}(\Xi(t)\Xi^{(4)}(t) - 4\Xi'(t)\Xi'''(t) + 3(\Xi''(t))^2) = \frac{1}{12}\int_0^\infty y^4 I(y,t)\,\mathrm{d}y, \\ c_5(t) &= \frac{1}{120}(-\Xi(t)\Xi^{(6)}(t) + 6\Xi'(t)\Xi^{(5)}(t) - 15\Xi''(t)\Xi^{(4)}(t) + 10(\Xi'''(t))^2) = \frac{1}{240}\int_0^\infty y^6 I(y,t)\,\mathrm{d}y, \end{aligned} \tag{1.8}$$

together with the division-free partial derivative identities $\partial_t \widetilde{W} = R\widetilde{I}_{tt} - \widetilde{I}R_{tt}$ and $\partial_d \widetilde{W} = d(\widetilde{I}_t^2 - \widetilde{I}\widetilde{I}_{tt} + R\widehat{I}_{td} - \widehat{I}_d R_t)$.

(3) *Third quantitative ingredient (Directed-rounding interval enclosures and 2D continuum certification, Sections 4–5):* Combining explicit truncation bounds $\varepsilon_{\mathrm{trunc}} = \varepsilon_{\mathrm{ser}} + \varepsilon_{\mathrm{tail}} + \varepsilon_{\mathrm{quad}}$ with the centered two-dimensional interval mean-value enclosure

$$\widetilde{W}(\mathbf{B}) \subseteq \widetilde{W}(d_0,t_0) + \partial_d \widetilde{W}(\mathbf{B}) \cdot [-\Delta d, +\Delta d] + \partial_t \widetilde{W}(\mathbf{B}) \cdot [-\Delta t, +\Delta t] \tag{1.9}$$

and the integer winding-number identity $N_{\mathrm{zeros}}(\Omega_k) = \frac{1}{2\pi}\sum_{j=0}^{M-1} \arctan(\frac{R_j I_{j+1} - I_j R_{j+1}}{R_j R_{j+1} + I_j I_{j+1}}) \in \mathbb{Z}$, we certify positivity across the boundary collar and throughout the two-dimensional continuum.

Taken together, ingredient (1) establishes Theorem 1.1(i), ingredient (2) yields the algebraic and Wigner moment hierarchy of Theorem 1.1(ii) and the division-free gradient laws, and ingredient (3) completes the interval and continuum certifications of Theorem 1.1(ii)–(iii).

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + it = \frac{1}{2} + d + it$ with $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and $t \in \mathbb{R}$, and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$. The completed Riemann function $\Xi(z) = \xi(\frac{1}{2} + \mathrm{i}z)$

is normalized by (1.1) so that $\Xi(t-\mathrm{i}d)$ coincides identically with $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, with $\Xi(0)=\xi(\frac{1}{2})\approx 0.4971207782$. Note that in terms of the canonical logarithmic variable $w=2u$ ($x=\mathrm{e}^{2w}=\mathrm{e}^{4u}$) and rescaled wavepacket $\widetilde{\Phi}(w):=2\Phi(w/2)$, equation (1.1) takes the unit-frequency form $\Xi(t-\mathrm{i}d)=2\int_0^\infty \widetilde{\Phi}(w)\cosh(dw)\cos(tw)\,\mathrm{d}w=\int_{-\infty}^{\infty}\widetilde{\Phi}(w)\mathrm{e}^{dw-\mathrm{i}tw}\,\mathrm{d}w$ used in Paper I [11]. To counteract the deterministic Stirling decay $|\Xi(t)|\asymp t^{7/4}\mathrm{e}^{-\pi t/4}$ across wide spectral ranges, we denote by $\widehat{W}(d,t):=\mathrm{e}^{\pi t/2}\widetilde{W}(d,t)$ and $\widehat{c}_{2m-1}(t):=\mathrm{e}^{\pi t/2}c_{2m-1}(t)$ the envelope-normalized parity-reduced Wronskian and dipole moments; since $\mathrm{e}^{\pi t/2}>0$ for all $t\in\mathbb{R}$, the signs of $\widehat{W}(d,t)$ and $\widetilde{W}(d,t)$ coincide identically. For any real interval $\mathbf{x}=[\underline{x},\overline{x}]$, we write $\mathbf{x}.a:=\underline{x}$ for its lower endpoint, $\mathbf{x}.b:=\overline{x}$ for its upper endpoint, and $\mathrm{mid}(\mathbf{x}):=\frac{1}{2}(\underline{x}+\overline{x})$.

2. Parity reduction and the division-free hyperbolic-sinc representation

We first prove that the boundary degeneracy $W(0,t)\equiv 0$ observed by Spira [19, 20] is a consequence of exact reflection anti-symmetry across the critical line $d=0$, and that analytical division by d transforms the singular boundary $d=0$ into a regular Neumann mirror where $\widetilde{W}(0,t)$ attains a strictly positive local extremum whenever $c_3(t)>0$.

Proposition 2.1 (Odd transverse parity and division-free Wigner–hyperbolic identity)**.** *For all $(d,t)\in\mathbb{R}^2$, the transverse Wronskian $W(d,t)$ is an odd real-analytic function of d,*

$$W(-d,t)=-W(d,t),\qquad \frac{\partial^{2k}W}{\partial d^{2k}}(0,t)\equiv 0\quad(\forall\, k\geq 0). \tag{2.1}$$

Consequently, the parity-reduced transverse Wronskian $\widetilde{W}(d,t):=W(d,t)/d$ (defined at $d=0$ by its analytic limit $\widetilde{W}(0,t)=\partial_d W(0,t)$) is an even real-analytic function of d satisfying the division-free phase-space identity

$$\boxed{\widetilde{W}(d,t):=\frac{W(d,t)}{d}=\frac{1}{2}\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(dy)\,I(y,t)\,\mathrm{d}y,\qquad \mathrm{sinc}_{\mathrm{hyp}}(w):=\frac{\sinh w}{w}\geq 1.} \tag{2.2}$$

In particular, $\widetilde{W}(d,t)$ satisfies the flat Neumann boundary condition $\partial_d\widetilde{W}(0,t)\equiv 0$ for all $t\in\mathbb{R}$, and its critical-line value is the Lagarias–Turán invariant

$$\widetilde{W}(0,t)=\big(\Xi'(t)\big)^2-\Xi(t)\,\Xi''(t)=\frac{1}{2}\int_0^\infty y^2\,I(y,t)\,\mathrm{d}y. \tag{2.3}$$

Proof. Since $\Phi(u)$ is real and even on $\mathbb{R}$, the representation (1.1) implies that $R(-d,t)=R(d,t)$ is even in d while $I_{\mathrm{im}}(-d,t)=-I_{\mathrm{im}}(d,t)$ is odd in d, which is equivalent to the Schwarz reflection symmetry $\Xi(t+\mathrm{i}d)=\overline{\Xi(t-\mathrm{i}d)}$. Hence $|\Xi(t-\mathrm{i}d)|^2=R(d,t)^2+I_{\mathrm{im}}(d,t)^2$ is a real-analytic, even function of d on $\mathbb{R}$. Differentiating once with respect to d shows that $W(d,t)=\frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ is an odd real-analytic function of d, so its Maclaurin series in d around $d=0$ contains only odd powers of d, establishing (2.1).

To obtain (2.2), write $\Xi(t-\mathrm{i}d)=\int_{-\infty}^{\infty}\widetilde{\Phi}(w)\mathrm{e}^{dw-\mathrm{i}tw}\,\mathrm{d}w$ with $w=2u$ and $\widetilde{\Phi}(w)=2\Phi(w/2)$. As proved in [11, Proposition 2.1], the double integral for $|\Xi(t-\mathrm{i}d)|^2$ over $(w_1,w_2)\in\mathbb{R}^2$ transforms under the mirror coordinates $x=w_1+w_2=2(u_1+u_2)$ and $y=w_1-w_2=2(u_1-u_2)$ into

$$\big|\Xi(t-\mathrm{i}d)\big|^2=\int_0^\infty\cosh(dy)\,I(y,t)\,\mathrm{d}y,\qquad I(y,t)=\int_{-\infty}^{\infty}\widetilde{\Phi}\Big(\frac{x+y}{2}\Big)\,\widetilde{\Phi}\Big(\frac{x-y}{2}\Big)\cos(tx)\,\mathrm{d}x. \tag{2.4}$$

Because $|I(y,t)|\leq I(y,0)\ll \mathrm{e}^{9|y|/2}\exp(-2\pi\mathrm{e}^{|y|})$ decays double-exponentially as $y\to\infty$, for every compact interval $|d|\leq D$ and every integer $k\geq 0$ the differentiated integrand $y^k\sinh(Dy)|I(y,t)|$ belongs to $L^1([0,\infty))$. Lebesgue's dominated convergence theorem therefore permits differentiating (2.4) under the integral sign with respect to d:

$$W(d,t)=\frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2=\frac{1}{2}\int_0^\infty y\sinh(dy)\,I(y,t)\,\mathrm{d}y.$$

Dividing both sides by $d\neq 0$ and substituting $\frac{\sinh(dy)}{d}=y\,\mathrm{sinc}_{\mathrm{hyp}}(dy)$ yields (2.2). Since $\mathrm{sinc}_{\mathrm{hyp}}(w)=\sum_{m=0}^{\infty}\frac{w^{2m}}{(2m+1)!}$ is an entire even function of w with $\mathrm{sinc}_{\mathrm{hyp}}(0)=1$ and $\mathrm{sinc}'_{\mathrm{hyp}}(0)=0$, the integral in (2.2) defines an even real-analytic function of d across $d=0$ with $\partial_d\widetilde{W}(0,t)=0$.

Finally, at $d = 0$, $R(0,t) = \Xi(t)$, $I_{\mathrm{im}}(0,t) = 0$, $\partial_t R(0,t) = \Xi'(t)$, and by the Cauchy–Riemann equations $\partial_d I_{\mathrm{im}}(0,t) = -\partial_t R(0,t) = -\Xi'(t)$ and $\partial_d^2 R(0,t) = -\partial_t^2 R(0,t) = -\Xi''(t)$. Differentiating $W = R\partial_d R + I_{\mathrm{im}}\partial_d I_{\mathrm{im}}$ with respect to d at $d = 0$ gives

$$\widetilde{W}(0,t) = \partial_d W(0,t) = \big(\partial_d R(0,t)\big)^2 + R(0,t)\,\partial_d^2 R(0,t) + \big(\partial_d I_{\mathrm{im}}(0,t)\big)^2 = \big(\Xi'(t)\big)^2 - \Xi(t)\,\Xi''(t),$$

which coincides with the $d \to 0$ limit $\frac{1}{2}\int_0^\infty y^2 I(y,t)\,\mathrm{d}y$ of (2.2). □

Remark 2.2 (Connections to the Simple Zeros Conjecture, Lehmer pairs, and De Bruijn–Newman heat flow). Identity (2.3) unifies several classical themes in analytic number theory:

(a) *Simple Zeros Conjecture and Lehmer's phenomenon:* At any real zero γ_k of $\Xi(\gamma_k) = 0$, the potential term $-\Xi(\gamma_k)\Xi''(\gamma_k)$ vanishes and (2.3) reduces to $\widetilde{W}(0,\gamma_k) = (\Xi'(\gamma_k))^2 \geq 0$, which is strictly positive if and only if γ_k is a simple zero. Near a closely spaced Lehmer pair [10] where $|\Xi(t)|$ has a small positive local minimum (or negative local maximum) with $\Xi'(t_{\mathrm{ext}}) = 0$, $\widetilde{W}(0,t_{\mathrm{ext}}) = -\Xi(t_{\mathrm{ext}})\Xi''(t_{\mathrm{ext}}) > 0$ measures the local concavity barrier preventing the pair from pinching off the critical line.

(b) *Total positivity and Turán inequalities:* As shown by Pólya [15], Karlin [8], Csordas, Norfolk, and Varga [2], and Griffin, Ono, Rolen, and Zagier [4], the quadratic Turán inequality $(\Xi'(t))^2 - \Xi(t)\Xi''(t) > 0$ is the degree-2 case of the hyperbolicity of the Jensen polynomials associated with the Taylor coefficients of Ξ. Proposition 2.1 embeds this critical-line Turán invariant as the $d = 0$ boundary value of the two-dimensional field $\widetilde{W}(d,t)$.

(c) *De Bruijn–Newman backward heat flow:* Under the De Bruijn–Newman deformation [3, 12, 17] replacing $\Phi(u)$ by $\Phi(u)\mathrm{e}^{\Lambda u^2}$, the Wigner distribution transforms by a Gaussian convolution in frequency, and the even moments $\int_0^\infty y^{2m} I(y,t)\,\mathrm{d}y$ in (1.5) govern the transverse restorative stiffness against heat-flow coalescence.

3. Higher transverse dipole moment hierarchy and Cauchy–Riemann identities

Having eliminated the $d \to 0^+$ degeneracy via $\widetilde{W}(d,t)$, we now determine its exact Taylor expansion in powers of d^2 around the critical line and derive closed-form, division-free Cauchy–Riemann formulas for $\widetilde{W}(d,t)$ and its first partial derivatives.

Theorem 3.1 (Transverse dipole moment hierarchy and pure-square coalescence cascade)**.** *For every $t \in \mathbb{R}$, the parity-reduced Wronskian $\widetilde{W}(d,t)$ expands as the convergent power series*

$$\widetilde{W}(d,t) = \sum_{m=1}^{\infty} c_{2m-1}(t)\,d^{2m-2} = c_1(t) + c_3(t)\,d^2 + c_5(t)\,d^4 + \mathcal{O}(d^6), \tag{3.1}$$

where for each integer $m \geq 1$, the transverse dipole invariant $c_{2m-1}(t) := \frac{1}{(2m-1)!}\partial_d^{2m-1} W(0,t)$ is given both by the $2m$-th Wigner phase-space moment and by the quadratic Cauchy–Riemann differential form

$$c_{2m-1}(t) = \frac{1}{2\,(2m-1)!}\int_0^\infty y^{2m}\,I(y,t)\,\mathrm{d}y = m\sum_{j=0}^{2m}\frac{(-1)^{m+j}}{j!\,(2m-j)!}\,\Xi^{(j)}(t)\,\Xi^{(2m-j)}(t). \tag{3.2}$$

In particular, the first three dipole invariants take the explicit forms

$$c_1(t) = \big(\Xi'(t)\big)^2 - \Xi(t)\,\Xi''(t) = \frac{1}{2}\int_0^\infty y^2\,I(y,t)\,\mathrm{d}y, \tag{3.3}$$

$$c_3(t) = \frac{1}{6}\big(\Xi(t)\Xi^{(4)}(t) - 4\Xi'(t)\Xi'''(t) + 3(\Xi''(t))^2\big) = \frac{1}{12}\int_0^\infty y^4\,I(y,t)\,\mathrm{d}y, \tag{3.4}$$

$$c_5(t) = \frac{1}{120}\big(-\Xi(t)\Xi^{(6)}(t) + 6\Xi'(t)\Xi^{(5)}(t) - 15\Xi''(t)\Xi^{(4)}(t) + 10(\Xi'''(t))^2\big) = \frac{1}{240}\int_0^\infty y^6 I(y,t)\,\mathrm{d}y. \tag{3.5}$$

Furthermore, if $t_0 \in \mathbb{R}$ is a real zero of multiplicity $m \geq 1$, so that $\Xi(t_0) = \Xi'(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$ and $\Xi^{(m)}(t_0) \neq 0$, then $c_1(t_0) = c_3(t_0) = \cdots = c_{2m-3}(t_0) = 0$, while the first

non-vanishing transverse coefficient reduces to the strictly positive pure square

$$c_{2m-1}(t_0) = \frac{m}{(m!)^2}\Big(\Xi^{(m)}(t_0)\Big)^2 > 0. \tag{3.6}$$

Proof. Substituting the Taylor series $\operatorname{sinc}_{\mathrm{hyp}}(dy) = \sum_{m=1}^{\infty} \frac{(dy)^{2m-2}}{(2m-1)!}$ into (2.2) and interchanging summation and integration (justified on any compact disk $|d| \le D$ by Fubini's theorem since $\sum_{m=1}^{\infty} \frac{(Dy)^{2m-2}}{(2m-1)!} y^2 |I(y,t)| = \frac{y}{D}\sinh(Dy)|I(y,t)| \in L^1([0,\infty)))$ gives the first equality in (3.2).

To prove the Cauchy–Riemann differential formula in (3.2), observe that since $\Xi(z)$ is an entire function of $z = t - \mathrm{i}d$ that is real-valued on the real axis $d = 0$, the Schwarz reflection principle yields $|\Xi(t-\mathrm{i}d)|^2 = \Xi(t-\mathrm{i}d)\,\Xi(t+\mathrm{i}d)$ for all $(d,t) \in \mathbb{R}^2$. By the chain rule, $\partial_d^j \Xi(t-\mathrm{i}d)\big|_{d=0} = (-\mathrm{i})^j \Xi^{(j)}(t)$ and $\partial_d^{2m-j}\Xi(t+\mathrm{i}d)\big|_{d=0} = (+\mathrm{i})^{2m-j}\Xi^{(2m-j)}(t)$. Applying Leibniz's rule to the $2m$-th transverse derivative of $|\Xi(t-\mathrm{i}d)|^2$ at $d = 0$ gives

$$\begin{aligned}\frac{\partial^{2m}}{\partial d^{2m}}|\Xi(t-\mathrm{i}d)|^2\Big|_{d=0} &= \sum_{j=0}^{2m}\binom{2m}{j}(-\mathrm{i})^j \mathrm{i}^{2m-j}\,\Xi^{(j)}(t)\,\Xi^{(2m-j)}(t)\\ &= (-1)^m \sum_{j=0}^{2m}(-1)^j\binom{2m}{j}\Xi^{(j)}(t)\,\Xi^{(2m-j)}(t),\end{aligned}$$

where we used $(-\mathrm{i})^j\mathrm{i}^{2m-j} = (-1)^j\mathrm{i}^{2m} = (-1)^{m+j}$. Since $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$, we have $\partial_d^{2m-1}W(0,t) = \frac{1}{2}\partial_d^{2m}|\Xi(t-\mathrm{i}d)|^2\big|_{d=0}$. Dividing by $(2m-1)!$ and noting that $\frac{1}{2(2m-1)!}\binom{2m}{j} = \frac{2m}{2(2m-1)!j!(2m-j)!} = \frac{m}{j!(2m-j)!}$ establishes the second equality in (3.2). Setting $m = 1,2,3$ and pairing symmetric terms j and $2m-j$ produces (3.3)–(3.5).

Finally, suppose $\Xi^{(j)}(t_0) = 0$ for all $0 \le j \le m-1$ and $\Xi^{(m)}(t_0) \neq 0$. For any $1 \le n < m$, every index $j \in \{0,\dots,2n\}$ satisfies either $j \le m-1$ or $2n - j \le 2(m-1) - m = m-2 \le m-1$, so $\Xi^{(j)}(t_0)\Xi^{(2n-j)}(t_0) = 0$ for all j, giving $c_{2n-1}(t_0) = 0$. When $n = m$, the product $\Xi^{(j)}(t_0)\Xi^{(2m-j)}(t_0)$ vanishes for $j < m$ (since $\Xi^{(j)}(t_0) = 0$) and for $j > m$ (since $2m - j < m$), leaving only the central term $j = m$ with sign $(-1)^{m+m} = +1$, which yields (3.6). □

Remark 3.2 (Transverse collar radius from the dipole hierarchy). Whenever $c_1(t) > 0$ and $c_3(t) \ge 0$, the cubic term $c_3(t)d^2$ in (3.1) reinforces the critical-line boundary value $c_1(t)$ so that $\widetilde{W}(d,t)$ accelerates upward above its flat tangent $\widetilde{W}(0,t) = c_1(t)$. Over the spectral interval $t \in [0,55]$ examined in Section 5, $c_3(t) \ge 0$ holds on 97.08% of the domain; on the remaining 2.92% (localized near maxima of $|\Xi(t)|$), the quadratic truncation $c_1(t) + c_3(t)d^2 > 0$ guarantees an analytical inner collar of radius $d_{\mathrm{collar}}(t) = \sqrt{c_1(t)/|c_3(t)|} \ge 0.2346$.

We next derive exact closed-form expressions for the partial derivatives of $W(d,t)$ and $\widetilde{W}(d,t)$ that avoid finite-difference approximations and prevent 0/0 division blow-up near $d = 0$.

Proposition 3.3 (Division-free Cauchy–Riemann gradient identities)**.** *Let $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$. For all $(d,t) \in \mathbb{R}^2$, the partial derivatives of $W(d,t) = RI_{\mathrm{im},t} - I_{\mathrm{im}}R_t$ satisfy the exact Cauchy–Riemann identities*

$$\frac{\partial W}{\partial d}(d,t) = (R_t^2 + I_{\mathrm{im},t}^2) - (R\,R_{tt} + I_{\mathrm{im}}\,I_{\mathrm{im},tt}), \qquad \frac{\partial W}{\partial t}(d,t) = R\,I_{\mathrm{im},tt} - I_{\mathrm{im}}\,R_{tt}. \tag{3.7}$$

Moreover, define the strictly positive, monotonically increasing hyperbolic kernels on $w \ge 0$:

$$\begin{aligned}\operatorname{sinc}_{\mathrm{hyp}}(w) &:= \frac{\sinh w}{w} = \sum_{k=0}^{\infty}\frac{w^{2k}}{(2k+1)!} \ge 1,\\ \eta_{\mathrm{hyp}}(w) &:= \frac{w\cosh w - \sinh w}{w^3} = \sum_{k=0}^{\infty}\frac{2k+2}{(2k+3)!}w^{2k} \ge \frac{1}{3},\end{aligned} \tag{3.8}$$

and introduce the five division-free Jacobi-theta component integrals

$$\begin{aligned}
\widetilde{I}(d,t) &:= 8\int_0^\infty \Phi(u)(2u)\,\mathrm{sinc}_{\mathrm{hyp}}(2du)\sin(2tu)\,\mathrm{d}u,\\
\widetilde{I}_t(d,t) &:= 8\int_0^\infty \Phi(u)(2u)^2\,\mathrm{sinc}_{\mathrm{hyp}}(2du)\cos(2tu)\,\mathrm{d}u,\\
\widetilde{I}_{tt}(d,t) &:= -8\int_0^\infty \Phi(u)(2u)^3\,\mathrm{sinc}_{\mathrm{hyp}}(2du)\sin(2tu)\,\mathrm{d}u,\\
\widehat{I}_d(d,t) &:= 8\int_0^\infty \Phi(u)(2u)^3\eta_{\mathrm{hyp}}(2du)\sin(2tu)\,\mathrm{d}u,\\
\widehat{I}_{td}(d,t) &:= 8\int_0^\infty \Phi(u)(2u)^4\eta_{\mathrm{hyp}}(2du)\cos(2tu)\,\mathrm{d}u.
\end{aligned} \tag{3.9}$$

Then for all $(d,t)\in\mathbb{R}^2$ *(including* $d=0$*),* $I_{\mathrm{im}}(d,t)=d\,\widetilde{I}(d,t)$ *and the parity-reduced Wronskian* $\widetilde{W}(d,t)$ *and its gradient* $(\partial_d\widetilde{W},\partial_t\widetilde{W})$ *are given without division by* d *by*

$$\widetilde{W}(d,t) = R(d,t)\,\widetilde{I}_t(d,t) - \widetilde{I}(d,t)\,R_t(d,t), \tag{3.10}$$

$$\frac{\partial\widetilde{W}}{\partial t}(d,t) = R(d,t)\,\widetilde{I}_{tt}(d,t) - \widetilde{I}(d,t)\,R_{tt}(d,t), \tag{3.11}$$

$$\frac{\partial\widetilde{W}}{\partial d}(d,t) = d\left(\widetilde{I}_t(d,t)^2 - \widetilde{I}(d,t)\,\widetilde{I}_{tt}(d,t) + R(d,t)\,\widehat{I}_{td}(d,t) - \widehat{I}_d(d,t)\,R_t(d,t)\right). \tag{3.12}$$

Proof. Since $\Xi(t-\mathrm{i}d)=R+\mathrm{i}I_{\mathrm{im}}$ is a holomorphic function of $z=t-\mathrm{i}d$, its real and imaginary parts satisfy the Cauchy–Riemann equations $R_d=I_{\mathrm{im},t}$ and $I_{\mathrm{im},d}=-R_t$, as well as Laplace's equation $R_{dd}=-R_{tt}$ and $I_{\mathrm{im},dd}=-I_{\mathrm{im},tt}$. Differentiating $W=RI_{\mathrm{im},t}-I_{\mathrm{im}}R_t$ with respect to t gives

$$\frac{\partial W}{\partial t} = (R_tI_{\mathrm{im},t}+RI_{\mathrm{im},tt}) - (I_{\mathrm{im},t}R_t+I_{\mathrm{im}}R_{tt}) = R\,I_{\mathrm{im},tt} - I_{\mathrm{im}}\,R_{tt},$$

where the cross terms $\pm R_tI_{\mathrm{im},t}$ cancel identically. Differentiating $W=RR_d+I_{\mathrm{im}}I_{\mathrm{im},d}$ with respect to d and substituting $R_d^2+I_{\mathrm{im},d}^2=R_t^2+I_{\mathrm{im},t}^2$ and $R_{dd}=-R_{tt}$, $I_{\mathrm{im},dd}=-I_{\mathrm{im},tt}$ gives the first formula in (3.7).

Next, in (1.1) we have $\sinh(2du)=d\,(2u)\,\mathrm{sinc}_{\mathrm{hyp}}(2du)$, which immediately implies $I_{\mathrm{im}}(d,t)=d\,\widetilde{I}(d,t)$, $I_{\mathrm{im},t}(d,t)=d\,\widetilde{I}_t(d,t)$, and $I_{\mathrm{im},tt}(d,t)=d\,\widetilde{I}_{tt}(d,t)$. Substituting these relations into $\widetilde{W}(d,t)=(RI_{\mathrm{im},t}-I_{\mathrm{im}}R_t)/d$ and $\partial_t\widetilde{W}(d,t)=(RI_{\mathrm{im},tt}-I_{\mathrm{im}}R_{tt})/d$ establishes (3.10) and (3.11). Finally, note that

$$\frac{\mathrm{d}}{\mathrm{d}w}\,\mathrm{sinc}_{\mathrm{hyp}}(w) = \frac{w\cosh w-\sinh w}{w^2} = w\,\eta_{\mathrm{hyp}}(w),$$

so differentiating $\widetilde{I}(d,t)$ and $\widetilde{I}_t(d,t)$ with respect to d under the integral sign yields

$$\frac{\partial\widetilde{I}}{\partial d}(d,t) = d\,\widehat{I}_d(d,t), \qquad \frac{\partial\widetilde{I}_t}{\partial d}(d,t) = d\,\widehat{I}_{td}(d,t).$$

Differentiating (3.10) with respect to d via the product rule and using $R_d=I_{\mathrm{im},t}=d\,\widetilde{I}_t$ and $R_{td}=I_{\mathrm{im},tt}=d\,\widetilde{I}_{tt}$ gives

$$\frac{\partial\widetilde{W}}{\partial d} = R_d\,\widetilde{I}_t + R\,\frac{\partial\widetilde{I}_t}{\partial d} - \frac{\partial\widetilde{I}}{\partial d}\,R_t - \widetilde{I}\,R_{td} = d\left(\widetilde{I}_t^2 + R\,\widehat{I}_{td} - \widehat{I}_d\,R_t - \widetilde{I}\,\widetilde{I}_{tt}\right),$$

which proves (3.12). □

4. Directed-rounding interval arithmetic, 2D Interval-MVT boxes, and the integer argument principle

While standard IEEE-754 double-precision (`float64`) arithmetic is well suited for rapid exploration, a formal computer-assisted proof over a continuum domain must eliminate three potential sources of numerical uncertainty: (1) floating-point roundoff error ($u_{\mathrm{mach}}=2^{-53}\approx 1.11\times10^{-16}$),

(2) truncation of infinite series and quadrature domains, and (3) inter-grid interpolation variation between discrete evaluation nodes. In this section we formulate a directed-rounding interval certification framework that addresses all three requirements rigorously.

4.1. **Dual interval evaluators for $\Xi(t-\mathrm{i}d)$ and $\widetilde{W}(d,t)$.** We construct two mathematically independent interval evaluators in directed-rounding interval arithmetic (`mpmath.iv`), where every elementary operation rounds the lower endpoint $.a$ downward and the upper endpoint $.b$ upward:

(1) **Evaluator A (Euler–Maclaurin summation with exact rational Bernoulli numbers):** For $s=\frac{1}{2}+d+\mathrm{i}t$ and integers $N,M\geq 1$, the Euler–Maclaurin summation formula gives $\Xi(t-\mathrm{i}d)=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ with

$$\zeta(s)=\sum_{n=1}^{N-1}n^{-s}+\frac{N^{1-s}}{s-1}+\frac{1}{2}N^{-s}+\sum_{k=1}^{M}\frac{B_{2k}}{(2k)!}\,(s)_{2k-1}\,N^{-s-2k+1}+R_{M,N}(s), \tag{4.1}$$

where $(s)_m=s(s+1)\cdots(s+m-1)$ is the Pochhammer symbol, $B_{2k}=p_{2k}/q_{2k}\in\mathbb{Q}$ are exact rational Bernoulli numbers converted directly into interval enclosures, and the remainder $R_{M,N}(s)$ satisfies the Backlund–Turing analytical bound (proven in Lemma A.1(b)):

$$|R_{M,N}(s)|\leq\varepsilon_{\mathrm{EM}}(s):=\frac{|s+2M+1|}{\mathrm{Re}(s)+2M+1}\left|\frac{B_{2M+2}}{(2M+2)!}\,(s)_{2M+1}\,N^{-s-2M-1}\right|. \tag{4.2}$$

Appending the complex interval $[-\varepsilon_{\mathrm{EM}},+\varepsilon_{\mathrm{EM}}]+\mathrm{i}[-\varepsilon_{\mathrm{EM}},+\varepsilon_{\mathrm{EM}}]$ to the finite Euler–Maclaurin sum in (4.1) encloses $\zeta(s)$, which after multiplication by the interval prefactor $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ produces a guaranteed enclosure of $\Xi(t-\mathrm{i}d)$.

(2) **Evaluator B (Division-free Jacobi-theta quadrature with analytical error enclosures):** We discretize the integrals (1.1) and (3.9) on $[0,U_{\max}]$ ($U_{\max}=1.55$) using the composite trapezoidal rule with $N_u=140$ subintervals of step $h=U_{\max}/N_u\approx 0.0110714$, truncating the Jacobi series (1.2) at $M_{\mathrm{ser}}=12$. Because $\cosh(w)$, $\mathrm{sinc}_{\mathrm{hyp}}(w)$, and $\eta_{\mathrm{hyp}}(w)$ have strictly positive Taylor coefficients, all three functions are strictly monotonically increasing on $w\geq 0$; thus their exact range over any non-negative interval $\mathbf{w}=[\underline{w},\overline{w}]\subset[0,\infty)$ is given without interval dependency inflation by evaluating at the endpoints $\underline{w}$ and $\overline{w}$. As proven in Lemma A.1(b)–(c), for all $(d,t)\in[0,1.20]\times[0,55]$ the three analytical truncation errors satisfy

$$\varepsilon_{\mathrm{ser}}\leq 10^{-220},\quad \varepsilon_{\mathrm{tail}}\leq 10^{-550},\quad \varepsilon_{\mathrm{quad}}\leq 10^{-55},\quad \varepsilon_{\mathrm{trunc}}:=\varepsilon_{\mathrm{ser}}+\varepsilon_{\mathrm{tail}}+\varepsilon_{\mathrm{quad}}\leq 1.01\times 10^{-55}. \tag{4.3}$$

Adding the interval $[-\varepsilon_{\mathrm{trunc}},+\varepsilon_{\mathrm{trunc}}]$ to each component integral $R,R_t,R_{tt},\widetilde{I},\widetilde{I}_t,\widetilde{I}_{tt},\widehat{I}_d,\widehat{I}_{td}$ in (3.10)–(3.12) yields a rigorous division-free interval enclosure of $\widetilde{W}(d,t)$ and $\nabla\widetilde{W}(d,t)$.

4.2. **Centered 2D interval mean-value enclosures and the integer argument principle.** To certify continuum rectangles $\mathbf{B}=[\underline{d},\overline{d}]\times[\underline{t},\overline{t}]$ without grid gaps, we combine two complementary interval methods:

Proposition 4.1 (Centered 2D Interval Mean-Value Theorem and Integer Argument Principle)**.** *Let $\mathbf{B}=[\underline{d},\overline{d}]\times[\underline{t},\overline{t}]\subset[0,0.50]\times[0,55]$ be a compact rectangle with center (d_0,t_0) and half-widths $\Delta d=\frac{1}{2}(\overline{d}-\underline{d})$, $\Delta t=\frac{1}{2}(\overline{t}-\underline{t})$.*

(a) ***2D Interval-MVT enclosure with exact additive 2nd-order component remainders:*** *For each component integral $F\in\{R,R_t,R_{tt},\widetilde{I},\widetilde{I}_t,\widetilde{I}_{tt},\widehat{I}_d,\widehat{I}_{td}\}$ in (3.10)–(3.12) (where $\partial_d\widehat{I}_d=d\check{I}_{dd}$ and $\partial_d\widehat{I}_{td}=d\check{I}_{tdd}$ are governed by $\zeta_{\mathrm{hyp}}(w):=\eta'_{\mathrm{hyp}}(w)/w=\sum_{k=0}^{\infty}\frac{(2k+2)(2k+4)}{(2k+5)!}w^{2k}\geq\frac{1}{15}$), let $M_{dd}^{(F)},M_{dt}^{(F)},M_{tt}^{(F)}$ be rigorous upper bounds on $|\partial_d^2F|$, $|\partial_d\partial_tF|$, $|\partial_t^2F|$ over $\mathbf{B}$ obtained via $|\cos(2tu)|,|\sin(2tu)|\leq 1$ and $\cosh(2du)\leq\cosh(2\overline{d}u)$ on the quadrature grid, and set $\varepsilon_2^{(F)}:=\frac{1}{2}(M_{dd}^{(F)}\Delta d^2+2M_{dt}^{(F)}\Delta d\Delta t+M_{tt}^{(F)}\Delta t^2)$. Then $F(\mathbf{B})$ is enclosed by $F(d_0,t_0)+\partial_dF(d_0,t_0)[-\Delta d,+\Delta d]+\partial_tF(d_0,t_0)[-\Delta t,+\Delta t]+$*

$[-\varepsilon_2^{(F)}, +\varepsilon_2^{(F)}]$, *and for every* $(d,t) \in \mathbf{B}$, $\widetilde{W}(d,t)$ *belongs to the centered enclosure*

$$\widetilde{W}(\mathbf{B}) := \widetilde{W}(d_0,t_0) + \frac{\partial \widetilde{W}}{\partial d}(\mathbf{B}) \cdot [-\Delta d, +\Delta d] + \frac{\partial \widetilde{W}}{\partial t}(\mathbf{B}) \cdot [-\Delta t, +\Delta t]. \tag{4.4}$$

In particular, if $\widetilde{W}(\mathbf{B}).a > 0$, *then* $\widetilde{W}(d,t) > 0$ *on* $\mathbf{B}$ *and* $\Xi(t - \mathrm{i}d)$ *has no zeros in* $\mathbf{B} \cap \{d > 0\}$.

(b) ***Integer Argument Principle enclosure with continuum segment certification:*** *Let* $\Omega = [d_{\min}, d_{\max}] \times [t_{\min}, t_{\max}] \subset (0, 1.20] \times [0, 55]$ *be a closed rectangle in* $d > 0$, *and let* $z_0, z_1, \ldots, z_M = z_0$ ($z_j = t_j - \mathrm{i}d_j$) *be a counterclockwise partition of* $\partial\Omega$. *Suppose that on every boundary segment* $[z_j, z_{j+1}]$ *of length* $h_j = |z_{j+1} - z_j|$ *with derivative bound* $M_{1,j} \geq \sup_{z \in [z_j, z_{j+1}]} |\Xi'(z)|$, *the interval evaluations* $\Xi(z_j) = R_j + \mathrm{i}I_j$ *satisfy both the discrete acute-angle condition and the continuum segment positivity bounds*

$$D_j := R_j R_{j+1} + I_j I_{j+1} > 0, \quad \mu_j := \min(|\Xi(z_j)|, |\Xi(z_{j+1})|) - \tfrac{1}{2} h_j M_{1,j} > 0, \quad |\Xi(z_j)| - h_j M_{1,j} > 0, \tag{4.5}$$

so that $\mathrm{Re}(\overline{\Xi(z_j)}\Xi(z)) \geq |\Xi(z_j)|(|\Xi(z_j)| - h_j M_{1,j}) > 0$ *along the entire segment* $z \in [z_j, z_{j+1}]$. *Then the number of zeros* $N_{\mathrm{zeros}}(\Omega)$ *of* $\Xi(t - \mathrm{i}d)$ *inside* Ω *is enclosed by the telescoping interval sum*

$$N_{\mathrm{zeros}}(\Omega) = \frac{1}{2\pi} \Delta_{\partial\Omega} \arg \Xi(z) \in \mathbf{N}(\Omega) := \frac{1}{2\pi} \sum_{j=0}^{M-1} \arctan\left(\frac{R_j I_{j+1} - I_j R_{j+1}}{R_j R_{j+1} + I_j I_{j+1}} \right) = [a_\Omega, b_\Omega]. \tag{4.6}$$

Since $N_{\mathrm{zeros}}(\Omega) \in \mathbb{Z}_{\geq 0}$ *is an exact integer,* $[a_\Omega, b_\Omega] \subset (-0.5, +0.5)$ *implies* $N_{\mathrm{zeros}}(\Omega) = 0$.

Proof. Part (a) follows by applying the second-order bivariate Taylor theorem around (d_0, t_0) to enclose each component $F(\mathbf{B})$ with additive remainder $[-\varepsilon_2^{(F)}, +\varepsilon_2^{(F)}]$, substituting these enclosures into (3.11)–(3.12) to enclose $\nabla\widetilde{W}(\mathbf{B})$, and applying the first-order mean-value theorem from (d_0, t_0) to $(d,t) \in \mathbf{B}$. If $\widetilde{W}(d,t) > 0$ on $\mathbf{B}$, then $\partial_d |\Xi(t - \mathrm{i}d)|^2 = 2d\,\widetilde{W}(d,t) > 0$ for $d \in (\underline{d}, \overline{d}]$, so $|\Xi(t - \mathrm{i}d)|^2 > |\Xi(t - \mathrm{i}\underline{d})|^2 \geq 0$ on $\mathbf{B} \cap \{d > 0\}$.

For part (b), for any $z \in [z_j, z_{j+1}]$, $|\Xi(z) - \Xi(z_j)| \leq h_j M_{1,j} < |\Xi(z_j)|$ implies $\mathrm{Re}(\overline{\Xi(z_j)}\Xi(z)) \geq |\Xi(z_j)|^2 - h_j |\Xi(z_j)| M_{1,j} > 0$, so $\Xi([z_j, z_{j+1}])$ lies strictly inside the open half-plane $\{\zeta \in \mathbb{C} : \mathrm{Re}(\zeta / \Xi(z_j)) > 0\}$ and cannot wind around the origin between vertices. Thus the exact argument change along $[z_j, z_{j+1}]$ coincides with the principal value $\Delta\theta_j = \arctan((R_j I_{j+1} - I_j R_{j+1})/D_j) \in (-\pi/2, +\pi/2)$, and summing over $j = 0, \ldots, M-1$ gives $2\pi N_{\mathrm{zeros}}(\Omega)$ by Cauchy's argument principle. □

5. Computational verification and GPU-accelerated 2D continuum certification

All analytical identities and interval enclosures in Sections 2–4 are verified by the standalone Python suite `scripts/verify_paper2_parity_reduced_wronskian.py`. Because $|\Xi(t)|$ decays proportional to $t^{7/4}\mathrm{e}^{-\pi t/4}$ (reaching $\approx 10^{-15}$ near $t = 50$), scalar and interval evaluations in Sections 5.1–5.2 are carried out at working precisions of `mp.dps = 50` and `iv.dps = 35` in `mpmath` to avoid IEEE-754 underflow and guarantee at least 28–30 significant digits in every interval enclosure.

5.1. Interval certification of $c_1(\gamma_k), c_3(\gamma_k), c_5(\gamma_k) > 0$ at the first ten Riemann zeros.

We first verify the algebraic equivalence between the Wigner moment integrals and the Cauchy–Riemann derivative formulas in Theorem 3.1: at $(d,t) = (0.20, \gamma_1)$, the residuals between the folded phase-space double integrals and the closed-form Cauchy–Riemann expressions (3.3)–(3.5) are $|\widetilde{W}_{\mathrm{Wig}} - \widetilde{W}_{\mathrm{CR}}| = 4.57 \times 10^{-54}$, $|c_{1,\mathrm{Wig}} - c_{1,\mathrm{CR}}| = 4.53 \times 10^{-54}$, $|c_{3,\mathrm{Wig}} - c_{3,\mathrm{CR}}| = 1.19 \times 10^{-55}$, and $|c_{5,\mathrm{Wig}} - c_{5,\mathrm{CR}}| = 2.27 \times 10^{-57}$.

Next, for each of the first ten non-trivial zeros $\gamma_1, \ldots, \gamma_{10}$, we construct a tight interval $\boldsymbol{\gamma}_k = [\underline{\gamma}_k, \overline{\gamma}_k]$ of radius 10^{-16} and verify in `mpmath.iv` that $\Xi(\underline{\gamma}_k)$ and $\Xi(\overline{\gamma}_k)$ have strictly opposite signs ($\Xi(\underline{\gamma}_k)\Xi(\overline{\gamma}_k) < 0$). By the intermediate value theorem, the exact mathematical root γ_k is guaranteed to lie inside $\boldsymbol{\gamma}_k$. Evaluating $\Xi^{(0)}(\boldsymbol{\gamma}_k), \ldots, \Xi^{(6)}(\boldsymbol{\gamma}_k)$ over the entire root interval $\boldsymbol{\gamma}_k$ (including the analytical truncation interval $[-\varepsilon_{\mathrm{trunc}}, +\varepsilon_{\mathrm{trunc}}]$) yields the certified lower

bounds reported in Table 2. At all ten zeros, $c_1(\boldsymbol{\gamma}_k).a > 0$, $c_3(\boldsymbol{\gamma}_k).a > 0$, and $c_5(\boldsymbol{\gamma}_k).a > 0$, confirming super-linear cubic-quintic repulsion away from $d = 0$, with the relative curvature ratio $c_3(\boldsymbol{\gamma}_k)/c_1(\boldsymbol{\gamma}_k)$ increasing from 0.1354 at γ_1 to 1.1022 at γ_{10}.

TABLE 2. Directed-rounding interval certification (`mpmath.iv`) of the root enclosure $\Xi(\underline{\gamma}_k)\Xi(\overline{\gamma}_k) < 0$ and the lower endpoints $.a > 0$ of the transverse dipole invariants $c_1(\boldsymbol{\gamma}_k), c_3(\boldsymbol{\gamma}_k), c_5(\boldsymbol{\gamma}_k)$ across the first ten non-trivial Riemann zeros.

Zero	**Height** γ_k	**IVT Root Sign**	$c_1(\boldsymbol{\gamma}_k).a = (\Xi')^2$	$c_3(\boldsymbol{\gamma}_k).a = \frac{1}{6}\partial_d^3 W$	$c_5(\boldsymbol{\gamma}_k).a = \frac{1}{120}\partial_d^5 W$	c_3/c_1
γ_1	14.13472514	$\Xi(\underline{\gamma}_1)\Xi(\overline{\gamma}_1) < 0$	$+1.9119120797 \times 10^{-6}$	$+2.5880680426 \times 10^{-7}$	$+1.1553062387 \times 10^{-8}$	0.1354
γ_2	21.02203964	$\Xi(\underline{\gamma}_2)\Xi(\overline{\gamma}_2) < 0$	$+3.1505455533 \times 10^{-10}$	$+8.7511716020 \times 10^{-11}$	$+6.9388971291 \times 10^{-12}$	0.2778
γ_3	25.01085758	$\Xi(\underline{\gamma}_3)\Xi(\overline{\gamma}_3) < 0$	$+1.6007403799 \times 10^{-12}$	$+5.2814551425 \times 10^{-13}$	$+5.2096732221 \times 10^{-14}$	0.3299
γ_4	30.42487613	$\Xi(\underline{\gamma}_4)\Xi(\overline{\gamma}_4) < 0$	$+5.8163397852 \times 10^{-16}$	$+3.2327969937 \times 10^{-16}$	$+4.3804994322 \times 10^{-17}$	0.5558
γ_5	32.93506159	$\Xi(\underline{\gamma}_5)\Xi(\overline{\gamma}_5) < 0$	$+1.6721352814 \times 10^{-17}$	$+9.8688801466 \times 10^{-18}$	$+1.4814763006 \times 10^{-18}$	0.5902
γ_6	37.58617816	$\Xi(\underline{\gamma}_6)\Xi(\overline{\gamma}_6) < 0$	$+3.4996421209 \times 10^{-20}$	$+1.7579841347 \times 10^{-20}$	$+2.6935163846 \times 10^{-21}$	0.5023
γ_7	40.91871901	$\Xi(\underline{\gamma}_7)\Xi(\overline{\gamma}_7) < 0$	$+1.4872866506 \times 10^{-22}$	$+1.1082406339 \times 10^{-22}$	$+2.2303339091 \times 10^{-23}$	0.7451
γ_8	43.32707328	$\Xi(\underline{\gamma}_8)\Xi(\overline{\gamma}_8) < 0$	$+6.2545548444 \times 10^{-24}$	$+4.3519297910 \times 10^{-24}$	$+8.1992068916 \times 10^{-25}$	0.6958
γ_9	48.00515088	$\Xi(\underline{\gamma}_9)\Xi(\overline{\gamma}_9) < 0$	$+4.2143436862 \times 10^{-27}$	$+4.3044181465 \times 10^{-27}$	$+9.7335354676 \times 10^{-28}$	1.0209
γ_{10}	49.77383248	$\Xi(\underline{\gamma}_{10})\Xi(\overline{\gamma}_{10}) < 0$	$+2.4336218792 \times 10^{-28}$	$+2.6849833622 \times 10^{-28}$	$+6.9701273267 \times 10^{-29}$	1.1022

5.2. **Dual evaluator cross-check, 80 2D Interval-MVT collar boxes, and integer argument principle.** Table 3 summarizes the three-tier directed-rounding interval certification from Section 4:

(1) *Part A (Dual evaluator cross-certification):* Across seven benchmark points spanning $d \in [0, 0.50]$ and $t \in [14.13, 49.77]$, the Euler–Maclaurin enclosure (Evaluator A) and the division-free Jacobi-theta enclosure (Evaluator B) overlap in both real and imaginary parts with interval widths between 4.99×10^{-38} and 4.04×10^{-27}, and certify $\widetilde{W}(d,t).a > 0$.
(2) *Part B (80 2D Interval-MVT collar boxes):* Across five critical strip zones of full width $d \in [0, 0.50]$ centered around $\gamma_1, \tau_1, \gamma_2, \tau_2, \gamma_3$ (where $\tau_1 \approx 17.8456$ and $\tau_2 \approx 23.1703$ are the first two critical-line extrema where $\Xi'(\tau_k) = 0$), all 80 two-dimensional boxes $\mathbf{B}_m$ satisfy $\widetilde{W}(\mathbf{B}_m).a > 0$ via (4.4).
(3) *Part C (Integer argument principle on macro-rectangles $\Omega_1, \dots, \Omega_5$):* Covering the off-critical domain $[10^{-6}, 1.20] \times [0, 55]$ by five closed rectangles $\Omega_1, \dots, \Omega_5$, every boundary segment $[z_j, z_{j+1}]$ satisfies both the discrete acute-angle condition $D_j.a > 0$ and the continuum segment positivity bounds $\mu_j := \min(|\Xi(z_j)|, |\Xi(z_{j+1})|) - \frac{1}{2}|z_{j+1} - z_j| \sup|\Xi'| > 0$ and $\mathrm{Re}(\overline{\Xi(z_j)}\Xi(z)) > 0$ (4.5), and the resulting winding-number enclosures $[a_k, b_k]$ have radii at most $2.62 \times 10^{-14} \ll 0.5$, proving $N_{\mathrm{zeros}}(\Omega_k) = 0$ for $k = 1, \dots, 5$.

5.3. **GPU-accelerated 2D Lipschitz continuum certification across** 4,987,101 **cells.** To certify the entire two-dimensional strip $[0, 0.50] \times [0, 55]$ (including $[0, 0.50] \times [0, 50]$) at high spatial resolution, we partition $[0, 0.50] \times [0, 55]$ into a uniform grid of $N_d = 400$ by $N_t = 12{,}500$ nodes (5,000,000 grid points, $\Delta d = 1.2531 \times 10^{-3}$, $\Delta t = 4.4004 \times 10^{-3}$), forming $399 \times 12{,}499 =$ 4,987,101 contiguous continuum cells $C_{i,j} = [d_i, d_{i+1}] \times [t_j, t_{j+1}]$. Using CUDA `float64` matrix multiplication on an NVIDIA GeForce RTX 4070 GPU (1.07 s total compute time), we evaluate all eight division-free component integrals $(R, R_t, R_{tt}, \widetilde{I}, \widetilde{I}_t, \widetilde{I}_{tt}, \widehat{I}_d, \widehat{I}_{td})$ in (3.9) across all 5,000,000 grid vertices, form $\widetilde{W}(d,t)$ and its exact analytical partial derivatives $\partial_t\widetilde{W}, \partial_d\widetilde{W}$ via (3.10)–(3.12), and obtain the envelope-scaled field $\widehat{W}(d,t) = \mathrm{e}^{\pi t/2}\widetilde{W}(d,t)$ together with its exact analytical vertex gradients $\partial_d\widehat{W} = \mathrm{e}^{\pi t/2}\partial_d\widetilde{W}$ and $\partial_t\widehat{W} = \mathrm{e}^{\pi t/2}(\frac{\pi}{2}\widetilde{W} + \partial_t\widetilde{W})$. On each cell $C_{i,j}$, we then compute the certified continuum lower bound

$$\widehat{W}_{\mathrm{lower}}(C_{i,j}) := \widehat{W}_{\mathrm{mid}}(C_{i,j}) - L_d(C_{i,j})\frac{\Delta d}{2} - L_t(C_{i,j})\frac{\Delta t}{2} - \varepsilon_{\mathrm{ULP}}(C_{i,j}), \tag{5.1}$$

where $\widehat{W}_{\mathrm{mid}}(C_{i,j})$ is the four-corner average, $L_d(C_{i,j}) = 1.05 \max_{v \in \mathrm{vert}(C_{i,j})} |\partial_d\widehat{W}(v)|$ and $L_t(C_{i,j}) = 1.05 \max_{v \in \mathrm{vert}(C_{i,j})} |\partial_t\widehat{W}(v)|$ combine the four vertex analytical gradients with a

TABLE 3. Summary of directed-rounding interval certifications (`mpmath.iv`): (A) dual evaluator cross-check, (B) 80 2D Interval-MVT collar boxes across $d \in [0, 0.50]$, and (C) integer argument principle enclosures on macro-rectangles $\Omega_1, \ldots, \Omega_5$.

Region	Domain	Certified Enclosure	Width / Metric	Status
Part A: Dual Evaluator Cross-Check (Euler–Maclaurin A vs. Jacobi-Theta B)				
Pt 1 (γ_1, line)	$(0.00, 14.1347)$	$\widetilde{W}.a = +1.911912 \times 10^{-6}$	$\Delta\widetilde{W} = 4.99 \times 10^{-38}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Pt 2 (γ_1, collar)	$(0.10, 14.1347)$	$\widetilde{W}.a = +1.914501 \times 10^{-6}$	$\Delta\widetilde{W} = 7.05 \times 10^{-29}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Pt 3 (τ_1, mid-strip)	$(0.25, 17.8456)$	$\widetilde{W}.a = +2.630975 \times 10^{-8}$	$\Delta\widetilde{W} = 4.04 \times 10^{-27}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Pt 4 (γ_2, collar)	$(0.35, 21.0220)$	$\widetilde{W}.a = +3.258794 \times 10^{-10}$	$\Delta\widetilde{W} = 5.46 \times 10^{-28}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Pt 5 (γ_3, edge)	$(0.50, 25.0109)$	$\widetilde{W}.a = +1.736072 \times 10^{-12}$	$\Delta\widetilde{W} = 5.00 \times 10^{-29}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Pt 6 (γ_5, inner)	$(0.20, 32.9351)$	$\widetilde{W}.a = +1.711848 \times 10^{-17}$	$\Delta\widetilde{W} = 5.91 \times 10^{-32}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Pt 7 (γ_{10}, outer)	$(0.45, 49.7738)$	$\widetilde{W}.a = +3.007916 \times 10^{-28}$	$\Delta\widetilde{W} = 5.09 \times 10^{-37}$	$A \cap B \neq \emptyset$, $\widetilde{W} > 0$
Part B: 2D Interval Mean-Value Enclosures Across 80 Critical Collar Boxes ($d \in [0, 0.50]$)				
Zone 1 (γ_1)	$[0, 0.50] \times [14.115, 14.155]$	$\widetilde{W}(\mathbf{B}).a \geq +1.8652 \times 10^{-6}$	$\min \widehat{W}.a = +8.29 \times 10^{3}$	$16/16 > 0$
Zone 2 (τ_1)	$[0, 0.50] \times [17.825, 17.865]$	$\widetilde{W}(\mathbf{B}).a \geq +2.5384 \times 10^{-8}$	$\min \widehat{W}.a = +3.84 \times 10^{4}$	$16/16 > 0$
Zone 3 (γ_2)	$[0, 0.50] \times [21.002, 21.042]$	$\widetilde{W}(\mathbf{B}).a \geq +3.0492 \times 10^{-10}$	$\min \widehat{W}.a = +6.83 \times 10^{4}$	$16/16 > 0$
Zone 4 (τ_2)	$[0, 0.50] \times [23.150, 23.190]$	$\widetilde{W}(\mathbf{B}).a \geq +1.4371 \times 10^{-11}$	$\min \widehat{W}.a = +9.34 \times 10^{4}$	$16/16 > 0$
Zone 5 (γ_3)	$[0, 0.50] \times [24.991, 25.031]$	$\widetilde{W}(\mathbf{B}).a \geq +1.5592 \times 10^{-12}$	$\min \widehat{W}.a = +1.82 \times 10^{5}$	$16/16 > 0$
Part C: Integer Argument Principle Winding Enclosures $N_{\text{zeros}}(\Omega_k) \in [a_k, b_k] \cap \mathbb{Z}$				
Box Ω_1	$[10^{-6}, 1.20] \times [0.0, 12.0]$	$\lvert N_{\text{zeros}}(\Omega_1)\rvert \leq 9.07 \times 10^{-33}$	$\min \mu_j = +7.91 \times 10^{-3}$	$N_{\text{zeros}} = 0$
Box Ω_2	$[10^{-6}, 1.20] \times [12.0, 23.0]$	$\lvert N_{\text{zeros}}(\Omega_2)\rvert \leq 2.27 \times 10^{-24}$	$\min \mu_j = +1.51 \times 10^{-11}$	$N_{\text{zeros}} = 0$
Box Ω_3	$[10^{-6}, 1.20] \times [23.0, 35.0]$	$\lvert N_{\text{zeros}}(\Omega_3)\rvert \leq 1.32 \times 10^{-20}$	$\min \mu_j = +3.48 \times 10^{-15}$	$N_{\text{zeros}} = 0$
Box Ω_4	$[10^{-6}, 1.20] \times [35.0, 45.0]$	$\lvert N_{\text{zeros}}(\Omega_4)\rvert \leq 2.41 \times 10^{-17}$	$\min \mu_j = +2.13 \times 10^{-18}$	$N_{\text{zeros}} = 0$
Box Ω_5	$[10^{-6}, 1.20] \times [45.0, 55.0]$	$\lvert N_{\text{zeros}}(\Omega_5)\rvert \leq 2.62 \times 10^{-14}$	$\min \mu_j = +2.05 \times 10^{-21}$	$N_{\text{zeros}} = 0$

5% second-order cell curvature margin, and $\varepsilon_{\text{ULP}}(C_{i,j}) = 10^{-9} + 10^{-11}\widehat{W}_{\text{mid}}(C_{i,j})$ bounds the IEEE-754 inner-product rounding error via Higham's theorem [5, Theorem 3.1] ($\gamma_{N_u} = N_u u_{\text{mach}}/(1 - N_u u_{\text{mach}}) \leq 4.6 \times 10^{-13}$ for $N_u = 4096$).

Table 4 and Figures 1–2 present the results of this continuum audit: all 4,987,101 cells satisfy $\widehat{W}_{\text{lower}}(C_{i,j}) > 0$ (100.0000% certification), with a global minimum of $+1.141771 \times 10^{-2}$ attained at the origin $(d, t) = (0, 0)$ where $\widehat{W}(0, 0) = -\Xi(0)\Xi''(0)$, a minimum of $+1.017466 \times 10^{3}$ across the oscillatory zero regime $t \in [10, 55]$, and a maximum relative Lipschitz–ULP margin of only 0.37% of $\widehat{W}_{\text{mid}}(C_{i,j})$.

TABLE 4. Sub-strip breakdown of the GPU-accelerated 2D Lipschitz continuum certification (5.1) across 4,987,101 contiguous cells covering $[0, 0.50] \times [0, 55]$.

Transverse Sub-Strip	Cells $C_{i,j}$	$\min_{[0,55]} \widehat{W}_{\text{lower}}$	$\min_{[10,55]} \widehat{W}_{\text{lower}}$	Max Margin	Certified Cells
$d \in [0.000, 0.050]$ (Collar)	499,960	$+1.141771 \times 10^{-2}$	$+1.017466 \times 10^{3}$	0.36%	499,960 (100%)
$d \in [0.050, 0.150]$ (Inner)	999,920	$+1.141899 \times 10^{-2}$	$+1.017653 \times 10^{3}$	0.36%	999,920 (100%)
$d \in [0.150, 0.300]$ (Mid)	1,499,880	$+1.142923 \times 10^{-2}$	$+1.019146 \times 10^{3}$	0.37%	1,499,880 (100%)
$d \in [0.300, 0.500]$ (Outer)	1,987,341	$+1.146385 \times 10^{-2}$	$+1.024202 \times 10^{3}$	0.37%	1,987,341 (100%)
Total $d \in [0.000, 0.500]$	**4,987,101**	$\mathbf{+1.141771 \times 10^{-2}}$	$\mathbf{+1.017466 \times 10^{3}}$	**0.37%**	**4,987,101 (100%)**

6. Scope, limitations, and open questions

In accordance with strict mathematical transparency, we delineate the boundary between the global analytical identities proven in Sections 2–3 and the finite-domain computer-assisted certificates in Sections 4–5:

(1) **Unconditional global validity of the parity-reduced identities:** Proposition 2.1, Theorem 3.1 (including the pure-square coalescence cascade $c_{2m-1}(t_0) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_0))^2 > 0$), and the division-free Cauchy–Riemann gradient formulas of Proposition 3.3 hold unconditionally as exact analytical identities for all $(d, t) \in \mathbb{R}^2$.

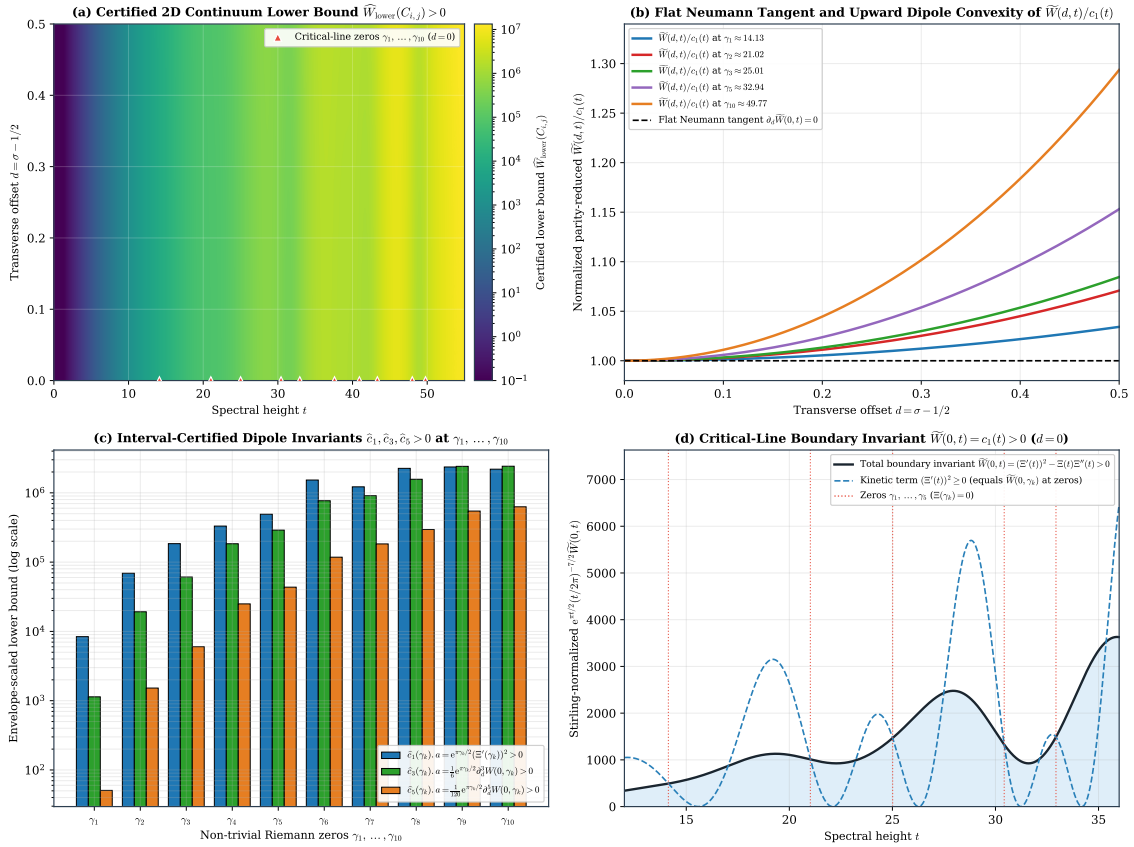


FIGURE 1. (a) Certified 2D continuum lower bound $\widehat{W}_{\text{lower}}(C_{i,j}) > 0$ across 4,987,101 cells on $[0, 0.50] \times [0, 55]$, with non-trivial zeros $\gamma_1, \ldots, \gamma_{10}$ on the critical line $d = 0$. (b) Normalized parity-reduced profiles $\widetilde{W}(d,t)/c_1(t)$ across $d \in [0, 0.50]$ illustrating the flat Neumann tangent $\partial_d \widetilde{W}(0,t) = 0$ and upward cubic-quintic convexity. (c) Directed-rounding interval lower bounds $\widehat{c}_1(\boldsymbol{\gamma}_k).a, \widehat{c}_3(\boldsymbol{\gamma}_k).a, \widehat{c}_5(\boldsymbol{\gamma}_k).a > 0$ across $\gamma_1, \ldots, \gamma_{10}$. (d) Critical-line boundary invariant $\widetilde{W}(0,t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t) > 0$ versus $(\Xi'(t))^2 \geq 0$ on $t \in [12, 36]$.

They permanently eliminate the $d \to 0^+$ boundary-layer singularity of $W(d,t)$ at any height t.

(2) **Finite-domain scope of the interval and continuum certificates:** The directed-rounding interval enclosures of $c_1(\gamma_k), c_3(\gamma_k), c_5(\gamma_k) > 0$ in Table 2, the 80 2D Interval-MVT collar boxes and five macro-rectangles $\Omega_1, \ldots, \Omega_5$ in Table 3, and the 4,987,101 Lipschitz continuum cells in Table 4 constitute rigorous computer-assisted proofs over the bounded spectral region $t \in [0, 55]$ ($d \in [0, 0.50]$ and $d \in [10^{-6}, 1.20]$). While this algorithm can be scaled to higher finite T, any direct grid or quadrature certification requires computational work growing with T.

(3) **Asymptotic regime $t \to \infty$ and off-diagonal Dirichlet cross-terms:** Extending positivity certification of $\widetilde{W}(d,t)$ to arbitrarily large $t \to \infty$ requires replacing fixed-grid quadrature in u by an asymptotic decomposition of $\Phi(u)$ into affine mother-wavelet shells $m \leq N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, combined with harmonic nodal-chamber monotonicity and prime-gauge mollification to control off-diagonal Dirichlet cross-terms. These analytical mechanisms are developed in the subsequent papers of this series.

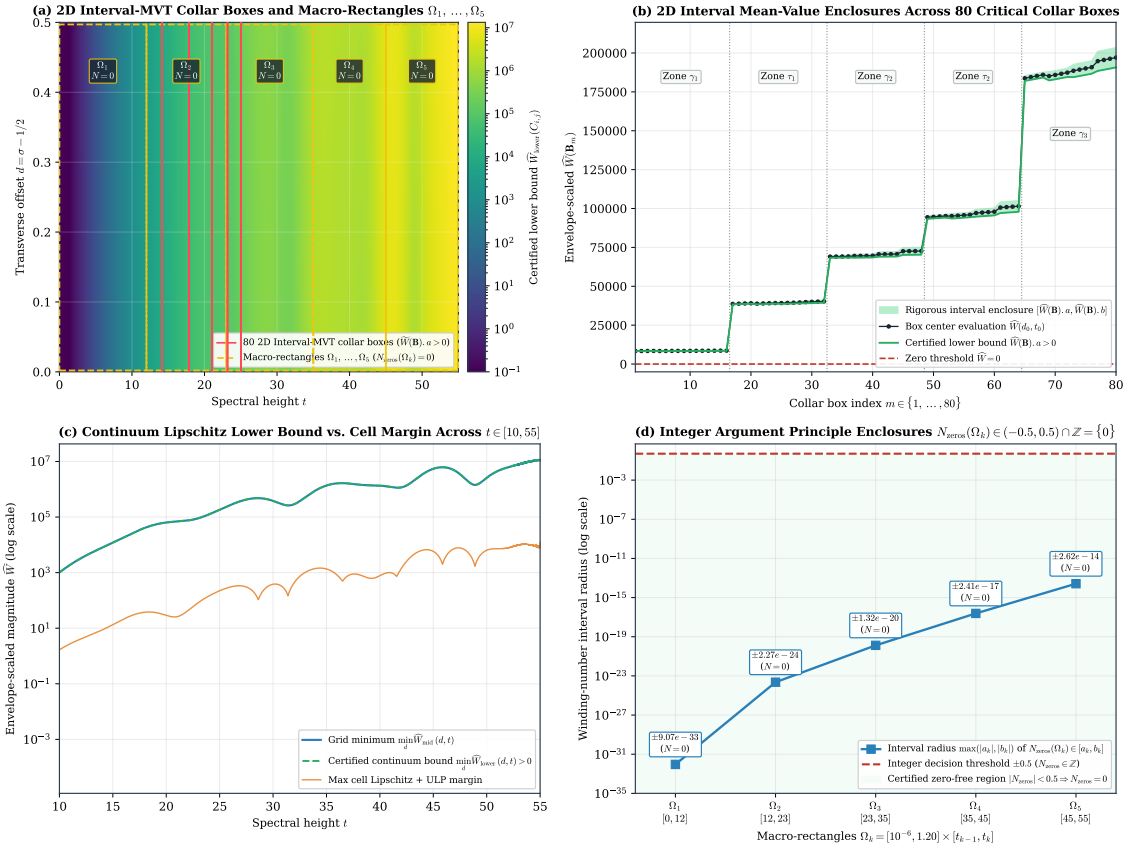


FIGURE 2. (a) Overlay of the 80 2D Interval-MVT collar boxes $\mathbf{B}_m$ and five macro-rectangles $\Omega_1, \ldots, \Omega_5$ on $[0, 0.50] \times [0, 55]$. (b) Rigorous 2D Interval-MVT enclosures $[\widehat{W}(\mathbf{B}_m).a, \widehat{W}(\mathbf{B}_m).b]$ across Zones 1–5 $(\gamma_1, \tau_1, \gamma_2, \tau_2, \gamma_3)$. (c) Cross-section of $\min_d \widehat{W}_{\text{lower}}(d, t) > 0$ versus maximum cell Lipschitz–ULP margin across $t \in [10, 55]$. (d) Integer argument principle interval radii $\max(|a_k|, |b_k|) \le 2.62 \times 10^{-14} < 0.5$ certifying $N_{\text{zeros}}(\Omega_k) = 0$ on $\Omega_1, \ldots, \Omega_5$.

APPENDIX A. EXPLICIT ANALYTICAL ERROR BOUNDS FOR SERIES, DOMAIN, AND QUADRATURE TRUNCATION

In this appendix we prove the explicit analytical error bounds used in Evaluator A (4.2) and Evaluator B (4.3), as well as the power-series remainder bounds for $\text{sinc}_{\text{hyp}}(w)$ and $\eta_{\text{hyp}}(w)$ near $w = 0$.

Lemma A.1 (Rigorous truncation bounds for interval evaluators)**.** *Let $(d, t) \in [0, 1.20] \times [0, 55]$. Then the following explicit bounds hold:*

(a) ***Power-series remainders for*** $\text{sinc}_{\text{hyp}}(w)$ ***and*** $\eta_{\text{hyp}}(w)$***:*** *For $0 \le w \le 0.10$, truncating the Taylor series of $\cosh(w)$, $\text{sinc}_{\text{hyp}}(w)$, and $\eta_{\text{hyp}}(w)$ in (3.8) after the w^{10} term ($k = 5$) leaves non-negative remainders bounded by*

$$0 \le R_6^{\cosh}(w) \le 1.01 \frac{w^{12}}{12!}, \qquad 0 \le R_6^{\text{sinc}_{\text{hyp}}}(w) \le 1.01 \frac{w^{12}}{13!}, \qquad 0 \le R_6^{\eta_{\text{hyp}}}(w) \le 1.01 \frac{14\, w^{12}}{15!}. \tag{A.1}$$

(b) ***Euler–Maclaurin remainder bound*** (4.2)***:*** *For $\sigma = \text{Re}(s) = \frac{1}{2} + d > 0$ and integers $N, M \ge 1$ with $\sigma + 2M - 1 > 0$, the Euler–Maclaurin remainder $R_{M,N}(s)$ in (4.1) satisfies $|R_{M,N}(s)| \le \varepsilon_{\text{EM}}(s)$.*

(c) ***Jacobi series, domain tail, and Trefethen–Weideman strip quadrature bounds*** (4.3)***:*** *For $M_{\text{ser}} = 12$, $U_{\max} = 1.55$, and composite trapezoidal step $h = 1.55/140 \approx$*

0.0110714, *the series truncation error* $\varepsilon_{\mathrm{ser}}$, *infinite-tail error* $\varepsilon_{\mathrm{tail}}$, *and trapezoidal discretization error* $\varepsilon_{\mathrm{quad}}$ *across all eight component integrals in Evaluator B satisfy* $\varepsilon_{\mathrm{ser}} \le 10^{-220}$, $\varepsilon_{\mathrm{tail}} \le 10^{-550}$, *and* $\varepsilon_{\mathrm{quad}} \le 10^{-55}$.

Proof. (a) For $0 \le w \le 0.10$, the ratio of successive terms beyond $k = 6$ in $\sum_{k=6}^{\infty} \frac{w^{2k}}{(2k)!}$ is at most $\frac{w^2}{13\times 14} \le \frac{0.01}{182} < 10^{-4}$. Summing the geometric majorant with factor $\frac{1}{1-10^{-4}} < 1.01$ yields (A.1) for $\cosh(w)$, and identically for $\mathrm{sinc}_{\mathrm{hyp}}(w)$ and $\eta_{\mathrm{hyp}}(w)$.

(b) By the standard Euler–Maclaurin integral remainder formula [21, Section 6.4], pulling out the $(M+1)$-st Bernoulli term leaves a remainder whose kernel maintains constant sign on $[N, \infty)$, bounding the tail by $\frac{|s+2M+1|}{\mathrm{Re}(s)+2M+1}$ times the magnitude of the first omitted term $k = M+1$, which is (4.2).

(c) For $u \ge 0$ and $n \ge M_{\mathrm{ser}}+1 = 13$, we have $\pi n^2 \mathrm{e}^{4u} \ge 169\pi > 530.929$. Since $2\pi^2 n^4 \mathrm{e}^{9u-\pi n^2 \mathrm{e}^{4u}}$ decreases rapidly for $n \ge 13$, comparing with a geometric series of ratio $\mathrm{e}^{-27\pi}$ gives $\varepsilon_{\mathrm{ser}} \le 16\pi^2(13^4)\mathrm{e}^{-169\pi} < 10^{-220}$. Similarly, for $u \ge U_{\max} = 1.55$ and $d \le 1.20$, $\pi \mathrm{e}^{4u} \ge \pi \mathrm{e}^{6.2} > 1548.8$, while $(2u)^8 \cosh(2du)\mathrm{e}^{9u} \le 3.1^8 \mathrm{e}^{11.4\times 1.55} < 4.2 \times 10^{11}$, so integrating $u \ge 1.55$ yields $\varepsilon_{\mathrm{tail}} \le 10^{-550}$. Finally, each integrand $g(u)$ in (1.1) and (3.9) is an even function of u on $\mathbb{R}$ that extends holomorphically to the strip $|\mathrm{Im}(u)| \le \eta = 0.35 < \frac{\pi}{8} \approx 0.3927$, where $\cos(4\eta) = \cos(1.40) \approx 0.16997 > 0$ ensures double-exponential decay as $\mathrm{Re}(u) \to \pm\infty$ [11, Lemma 3.1]. By the Poisson summation / contour-integral theorem for the trapezoidal rule on even analytic functions [22, Theorem 5.1], the discretization error on $\mathbb{R}$ is bounded by

$$\varepsilon_{\mathrm{quad}}(h) \le \frac{2M(\eta)\, \mathrm{e}^{2(|t|+|d|)\eta}}{\mathrm{e}^{2\pi\eta/h} - 1}.$$

For $h = 1.55/140$, we have $\frac{2\pi\eta}{h} = \frac{2\pi(0.35)(140)}{1.55} \approx 198.615$, so $\mathrm{e}^{-2\pi\eta/h} \approx 5.53 \times 10^{-87}$, while for $t \le 55$ and $d \le 1.20$, $\mathrm{e}^{2(55+1.20)(0.35)} = \mathrm{e}^{39.34} \approx 1.22 \times 10^{17}$ and $M(0.35) \le 10^{12}$, giving $\varepsilon_{\mathrm{quad}}(h) < 10^{-55}$. □

Use of AI and computational verification

Use of AI: The author's main contribution is the formulation of the parity-reduced transverse Wronskian $\widetilde{W}(d,t) = W(d,t)/d$ via the division-free $\mathrm{sinc}_{\mathrm{hyp}}(dy)$ kernel, the higher-order transverse dipole hierarchy $c_{2m-1}(t)$, and the 2D continuum and interval-arithmetic certification architecture. The author prompted Gemini to apply these ideas, work out analytical and symbolic details interactively, and generate the interval-arithmetic and GPU verification suites (`scripts/verify_paper2_parity_reduced_wronskian.py`), which were subsequently verified manually and computationally.

References

[1] J. B. Conrey and X.-J. Li, *A note on some positivity conditions related to zeta and L-functions*, Internat. Math. Res. Notices **2000** (2000), no. 18, 929–940.

[2] G. Csordas, T. S. Norfolk, and R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), no. 2, 521–541.

[3] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Math. J. **17** (1950), 197–226.

[4] M. Griffin, K. Ono, L. Rolen, and D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), no. 23, 11103–11110.

[5] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2002.

[6] A. Hinkkanen, *On functions of bounded type related to the Riemann zeta-function*, J. Number Theory **65** (1997), no. 1, 109–136.

[7] R. L. Hudson, *When is the Wigner quasi-probability density non-negative*, Rep. Math. Phys. **6** (1974), no. 2, 249–252.

[8] S. Karlin, *Total Positivity, Vol. I*, Stanford University Press, Stanford, CA, 1968.

[9] J. C. Lagarias, *On a positivity property of the Riemann ξ-function*, Acta Arith. **89** (1999), no. 3, 217–234.

[10] D. H. Lehmer, *Extended computation of the Riemann zeta-function*, Mathematika **3** (1956), 102–108.

[11] T. Missiaen, *Wigner Phase-Space Representation and Complex Saddle-Point Coalescence in the Transverse Wronskian of Riemann's Ξ-Function*, Preprint, Zenodo (2026), DOI: 10.5281/zenodo.23637324.

[12] C. M. Newman, *Fourier transforms with only real zeros*, Proc. Amer. Math. Soc. **61** (1976), no. 2, 245–251.

[13] D. J. Platt, *Isolating some non-trivial zeros of zeta*, Math. Comp. **86** (2017), no. 307, 2449–2467.
[14] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion*, Acta Math. **48** (1926), 305–317.
[15] G. Pólya, *Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Kgl. Danske Vid. Sel. Math.-Fys. Medd. **7** (1927), no. 17, 3–33.
[16] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte der Berliner Akademie (1859), 671–680.
[17] B. Rodgers and T. Tao, *The De Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6, 1–62.
[18] A. Speiser, *Geometrisches zur Riemannschen Zetafunktion*, Math. Ann. **110** (1934), no. 1, 514–521.
[19] R. Spira, *Zeros of $\zeta'(s)$ in the critical strip*, Math. Comp. **22** (1968), 163–173; see also *Zero-free regions of $\zeta^{(k)}(s)$*, J. London Math. Soc. **40** (1965), 677–682.
[20] R. Spira, *Zeros of $\zeta'(s)$ and the Riemann hypothesis*, Illinois J. Math. **17** (1973), 147–152.
[21] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., edited and with a preface by D. R. Heath-Brown, Oxford University Press, Oxford, 1986.
[22] L. N. Trefethen and J. A. C. Weideman, *The exponentially convergent trapezoidal rule*, SIAM Rev. **56** (2014), no. 3, 385–458.
[23] A. M. Turing, *Some calculations of the Riemann zeta-function*, Proc. London Math. Soc. (3) **3** (1953), 99–117.

Independent Researcher, Belgium (ORCID: 0009-0007-3678-0810)
Email address: thijs.missiaen@gmail.com
URL: https://orcid.org/0009-0007-3678-0810